Pandelis Zembashis

CTO | Go Runtime & Sandbox Systems | AI Agent Infrastructure

p@ndel.is | pa.ndel.is | github.com/PandelisZ | x.com/PandelisZ | linkedin.com/in/Pandelis

EXECUTIVE SUMMARY

CTO and hands-on systems engineer building sandboxed AI-agent infrastructure in Go. At Cosine, led the runtime and platform work behind model-controlled coding environments: a single-binary Go harness, tool execution layer, deterministic replay, multi-agent orchestration, Kubernetes/gVisor remote sandboxes, and secure enterprise deployments across cloud, VPC, on-prem, and air-gapped environments. Direct fit for xAI's Sandbox Service: safe model access to compute, code execution, tool use, remote environments, filesystem-heavy workloads, and cluster-backed agent training/product infrastructure.

PROFESSIONAL EXPERIENCE

Cosine | CTO | Oct. 2023 – Present

Applied AI lab building autonomous coding agents, sandboxed execution, and model-training infrastructure.

- Led product and engineering for Cosine's AI coding platform, including the Go runtime, tool execution layer, sandbox bridge, cloud-agent substrate, and enterprise deployment model.
- Built Cosine's own sandboxing path from 2024 onward: Kubernetes-scheduled remote agents, gVisor/runsc isolation, ephemeral per-session execution, network-policy boundaries, and secure shell/tool access for LLM-controlled coding tasks.
- Architected 'cos2' as a portable single Go harness powering CLI, desktop, VS Code/WebAssembly, API-server sessions, and remote Keef pods, giving every surface one runtime, one event model, and one release path.
- Designed event-bus orchestration for streaming tool calls, checkpoints, deterministic replay, subagent lifecycle events, memory writes, file locks, and reliability-focused debugging across local and remote sessions.
- Reworked filesystem and repository-ingestion hot paths around native Go concurrency, bundled ripgrep-class search, streaming file reads, and low-overhead tool results for large production codebases.
- Built the ML flywheel around the harness: structured rollout logs, test/LSP/checkpoint signals, human-review outcomes, and sandboxed parallel rollouts usable as reward and evaluation signals for coding agents.
- Supported private VPC, on-prem, and air-gapped deployments for regulated customers where code, model artifacts, execution logs, and tenant data must remain isolated and auditable.

11x.ai | Head of Engineering | Apr. 2023 – Sep. 2023

Fastest-growing UK AI startup from 0 to $1M ARR in under six months. TechCrunch coverage.

- Scaled the codebase and engineering organization behind Alice as 11x.ai grew from 0 to $1M ARR in under six months.
- Scaled outbound email and event-driven automation systems that processed millions of email chains and billions of tokens.
- Established engineering cadence, incident response habits, and onboarding rhythms during rapid seed-to-Series A growth.

Papaya | Technical Lead | Feb. 2022 – Mar. 2023

Marketplace and fleet platform helping electrify commercial and last-mile logistics. TechCrunch coverage.

- Led delivery of a fleet management platform for light electric vehicles in Next.js, React Query, and Tailwind CSS.
- Modernized the Ruby on Rails backend by introducing GraphQL APIs, improving product iteration across web, operations, and marketplace workflows.
- Established development cadences and product intake processes that improved execution across engineering and product.

Weezy (acquired by Getir) | Senior Software Engineer | Aug. 2021 – Jan. 2022

Rapid grocery delivery startup building warehouse, ordering, and customer systems.

- Took over the core ordering and processing monolith in Ruby on Rails, owning day-to-day delivery and forward planning during a high-growth phase.
- Built warehouse operations systems and supporting AWS and Heroku infrastructure with a focus on scale, availability, and security.

finleap connect | Senior Software Engineer | Nov. 2020 – Aug. 2021
White-label fintech platform powering digital banking products and apps.

- Built GraphQL APIs on NestJS and cross-platform clients in React Native, Apollo, Node.js, and TypeScript for a multi-tenant fintech platform.
- Led alerting and observability improvements while supporting migration from ECS to Amazon Lambda.
- Owned client-specific delivery across the stack, from GraphQL resolvers and mutations to customer-facing mobile screens.

IoT CONNCTD GmbH | Full Stack Engineer | Jul. 2019 – Nov. 2020
Cloud IoT platform unifying device protocols behind a GraphQL API.

- Built frontend interfaces, SDKs, and internal tooling for a cloud IoT platform exposing a unified GraphQL API across multiple device protocols.
- Designed authentication, update, and upstream synchronization flows using TypeScript, React, MobX, and Apollo GraphQL.

Substrakt | Software Developer | Aug. 2018 – Jun. 2019
Digital product agency building bespoke publishing, arts, and cultural web platforms.

- Built bespoke WordPress features, search experiences, and internal tooling for design-led client websites.
- Modernized the JavaScript toolchain from Gulp to Webpack, improving development and deployment workflows.

D4 Software | Software Engineer | Jun. 2017 – Jul. 2018
Product engineering consultancy building client platforms, tools, and cloud infrastructure.

- Built client-facing features and internal tools across product and platform engagements using JavaScript and .NET.
- Maintained AWS infrastructure, serverless services, databases, and virtual machines while modernizing the frontend stack with Webpack, TypeScript, and Vue.js.

WRITING & PUBLIC WORK

Why we built our own harness: Technical deep dive on Cosine's Go harness, gVisor-sandboxed Kubernetes substrate, tool execution, replay, multi-agent orchestration, and ML feedback loop.
Cosine Swarm: Long-horizon agent orchestration, parallel work execution, and coordination patterns for AI software engineering.
Cosine CLI runtime: Native search, local workflows, and agent execution improvements.

EDUCATION

Birmingham City University | Bachelor's Degree, Computer Science with Year in Industry | 2015 – 2019

AREAS OF EXPERTISE

Sandbox & Runtime Systems: Go, Linux, Kubernetes, gVisor/runsc, containers, remote execution, process isolation, WebSockets, protobuf, shell/tool control
Agent Infrastructure: Tool execution, deterministic replay, checkpointing, EventBus design, multi-agent orchestration, file locks, memory systems, browser/UI control
Performance & Systems: Filesystem traversal, repository indexing, native search, streaming I/O, concurrency, resource scheduling, cold-start optimization, observability
Production Platforms: AWS, VPC/on-prem/air-gapped deployment, RBAC/SSO, audit logging, encryption, network policy, CI/CD, incident-aware operations